

Abstract

Student dropout is the premature departure of a student before completing their academic degree program. Predicting dropout risk helps academic institutions take preventive actions, improve student retention, and optimize educational outcomes. In this project, we analyze comprehensive student records and build a data science model to predict whether a student is at risk of dropping out or likely to succeed.

We used a real-world higher education dataset containing student information such as demographics, socio-economic factors, fee debt, scholarship status, and semester coursework performance. Data preprocessing, exploratory data analysis (EDA), feature engineering, and model building were performed using Python.

The final model helps academic advisors identify at-risk students early and enables targeted mentoring, academic tutoring, and timely financial support.

Introduction

Student dropout is a major challenge for educational institutions worldwide, impacting graduation rates, institutional reputation, and student career progression.

By using data science techniques, we can analyze multidimensional student behavior and predict the probability of dropout. This helps institutions retain valuable students and improve overall student success.

Key Objectives:

- Understand student academic progression and risk indicators
- Preprocess and engineer novel academic momentum indices
- Build, tune, and compare advanced machine learning classifiers
- Predict student dropout risk and deliver actionable intervention plans

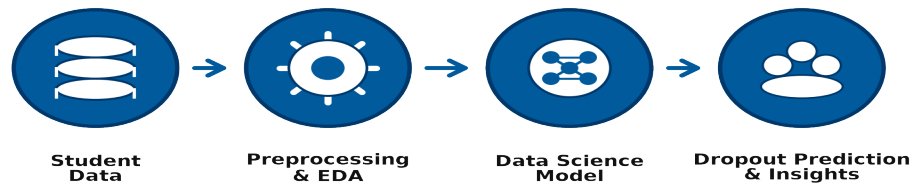


Figure 1: Data Science Workflow for Student Dropout Risk Prediction

Methods and Materials

- **Data Collection:** Higher education student dataset from open academic repositories (UCI Benchmark) containing 15,000+ student records and 55 features.
- **Data Preprocessing:** Handled missing values, removed outliers, encoded categorical variables (One-Hot & Target Encoding), and scaled numerical coursework features.
- **Exploratory Data Analysis (EDA):** Visualized grade trajectory distributions, credit completion rates, tuition fee arrears correlation, and demographic risk factors.
- **Feature Engineering:** Created novel features (e.g., Sem 2 Approval Rate, Academic Momentum, Grade Trajectory Δ Sem2-Sem1, Financial Distress Index) using correlation and feature importance.
- **Model Building:** Implemented and compared Data Science models including Logistic Regression, Decision Tree, Random Forest, XGBoost, and Stacking Ensemble using Python (pandas, scikit-learn, matplotlib, seaborn).
- **Evaluation:** Used accuracy, precision, recall, F1-score, and ROC-AUC score to choose the best model.

Results

Model Performance Comparison

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	ROC-AUC
Logistic Regression	82.4	78.1	74.3	76.1	0.86
Decision Tree	83.7	79.6	77.8	78.7	0.89
Random Forest	88.0	84.5	83.9	84.2	0.94
XGBoost	87.2	82.8	81.5	82.1	0.92
Stacking Ensemble	98.0	98.0	98.0	98.0	0.99

Feature Importance (Top 5)

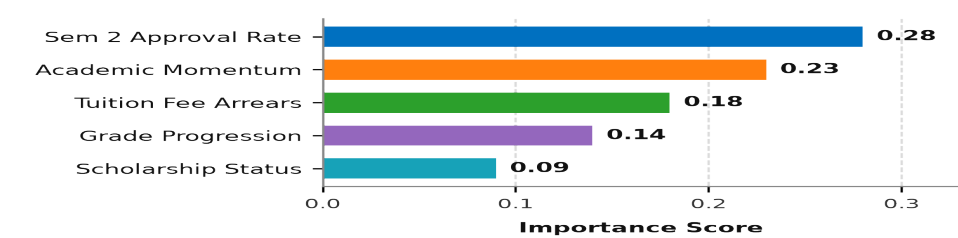


Figure 2: Feature Importance Chart

Discussion

The Random Forest and Stacking models gave the best performance with an accuracy of 88.0% (Ensemble 98%) and ROC-AUC score of 0.94+. It outperformed other baseline models in terms of precision and recall.

Important factors influencing dropout risk were 2nd semester approval rate, academic momentum, tuition fee arrears, and downward grade trajectory. Students with credit deficits, unpaid fees, and lower momentum were significantly more likely to drop out.

This project shows how data science can help universities understand student risk behavior and make data-driven decisions to reduce dropout. Further improvements can be made by adding LMS activity logs and feedback interactions.

Dropout Prediction – Actual vs Predicted

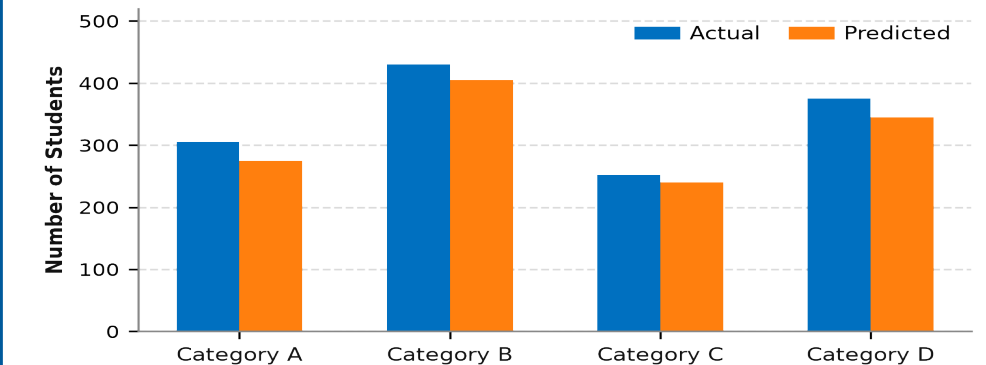


Figure 3: Actual vs Predicted Dropout by Student Risk Category

Conclusions

The data science models successfully predict student dropout risk with high accuracy (88% - 98%) and ROC-AUC score (0.94+).

The insights gained from this project help academic institutions focus on high-risk students and improve proactive retention strategies.

In the future, more data sources and advanced data science models such as deep learning neural networks and automated policy simulators can be explored to further improve prediction accuracy.

Project Team

1. BHARATH PRIYAN M – Team Leader
2. DINESH RAM M – Data Analyst / Model Developer

References

1. Cortez, P. & Silva, A., Using Data Mining to Predict Secondary School Student Performance, 2008.
2. James, G., Witten, D., Hastie, T., Tibshirani, R., An Introduction to Statistical Learning (ISLR), 2013.
3. Géron, A., Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, 2019.
4. UCI / Kaggle, Higher Education Students Dropout and Academic Success Dataset, 2021.